

MeCapture: Capturing and Visualizing Long-Term Body Changes with Mobile AR

Nhan (Nathan) Tran
Cornell University
Computer Science
Ithaca, New York, USA
nt322@cornell.edu

Ethan Yang
Cornell University
Computer Science
Ithaca, New York, USA
eey8@cornell.edu

Abe Davis
Cornell University
Computer Science
Ithaca, New York, USA
abedavis@cornell.edu

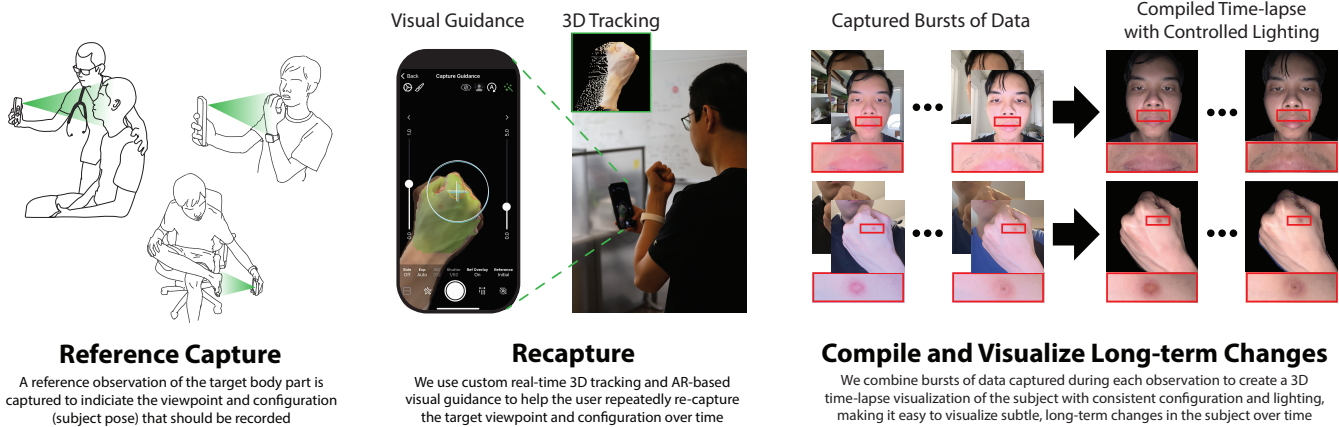


Figure 1: Our mobile app, MeCapture, creates Personal Time-Lapse (PTL) visualizations of long-term body changes. Starting with a reference observation (left), the app guides users during recapture (middle) to replicate the same viewpoint and body configuration. Computational illumination ensures consistent lighting across different environments (right), producing interactive 3D visualizations of the subject under controlled lighting.

Abstract

MeCapture is a mobile augmented reality (AR) application that enables users to document and visualize long-term bodily changes through Personal Time-Lapse (PTL). The app uses custom 3D tracking and AR-based visual guidance to help users repeatedly capture the same viewpoint and body configuration. Computational illumination ensures consistent lighting and color across different environments over time. In this demo, attendees will try the AR capture interface on various body parts like hands or face and interact with multiple 3D time-lapse visualizers showing real case studies, including healing progressions, facial hair growth, and plant growth. This work is based on our UIST 2024 publication [Tran et al. 2024]. The app is available for free on the App Store (<https://MeCapture.com>).

ACM Reference Format:

Nhan (Nathan) Tran, Ethan Yang, and Abe Davis. 2025. MeCapture: Capturing and Visualizing Long-Term Body Changes with Mobile AR. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Appy Hour (SIGGRAPH Appy Hour '25)*, August 10–14, 2025, Vancouver, BC, Canada.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
SIGGRAPH Appy Hour '25, Vancouver, BC, Canada
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1552-5/25/08
<https://doi.org/10.1145/3721260.3733982>

Canada. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3721260.3733982>

1 Introduction

Our bodies are constantly changing. Some changes are fast and temporary, like the motion involved in walking or breathing. Other changes are slower and more permanent, like those involved in growth and healing. These slower changes are important indicators of human health, but they are also some of the hardest to capture and visualize, as their short-term progress can be difficult to distinguish from transient changes of viewpoint, body pose, and lighting.

Consider a simple example: as a scraped knee begins to heal, the wounded area will shrink. However, the shrinkage observed in a day may be small compared to changes in size caused by simply bending the knee, which stretches or loosens skin around the kneecap. The color of the scrape may also change over time, but that can be difficult to distinguish from the effects of viewing the knee under different lighting conditions.

MeCapture addresses this challenge by helping users control the conditions of repeated observations taken over long periods of time. The app compiles these captured observations into an interactive visualization of long-term changes that we call Personal Time-Lapse.



Figure 2: Long-term results of Personal Time-Lapse using *MeCapture*. Case Study 1: Hair Growth Monitoring over 40 days (left), and Case Study 2: Wound Healing Tracking over 35 days (right). Top rows show captures with flash, second rows show captures without flash, and third rows show the consistent lighting results after computationally removing environmental lighting. Bottom rows provide detailed views of the tracked regions with changes over time.

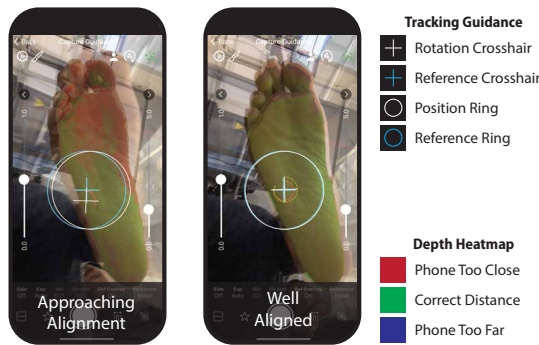


Figure 3: Body configuration guidance uses overlays and heatmaps, while viewpoint guidance uses crosshairs and rings.

2 How It Works

MeCapture guides users through three key steps. First, users capture a reference observation of their subject under desired conditions. The app then uses custom 3D tracking and AR-based visual guidance to help users repeatedly recapture the target viewpoint and body configuration over time. During capture, the app takes two photos in quick succession with and without flash, then computationally removes environmental lighting by taking their difference, creating consistent results whether taken indoors, outdoors, day or night.

As shown in Fig. 3, the AR guidance interface shows users an overlay of their reference image, depth heatmaps that indicate correct positioning (red for too close, blue for too far, green for correct), rotation crosshairs, and position rings. When alignment criteria are met, the app automatically triggers capture. A full video walk-through can be viewed at <https://youtu.be/75TkI7rdmmA>.

3 Appy Hour Demo Experience:

Attendees will experience *MeCapture* through hands-on interaction:

Try the Interface: Capture your own hand or face using our AR guidance system. Experience how the visual feedback helps achieve precise alignment that would be nearly impossible with a standard camera app.

Explore Real Results: Interact with multiple 3D time-lapse visualizers showing compelling case studies—facial hair growth, wound healing progressions, and other transformations captured by real users over weeks and months. Examples are in Figure 2 and interactive 3D visualization are available on our [project website](#).

See the Technology. Understand how custom 3D tracking works on non-rigid human subjects where standard mobile AR fails, and how computational lighting creates consistent results across different environments.

4 Applications and Impact

Our work has many possible downstream uses, with medical applications being our primary motivation: potentially enabling users to document wound healing, skin conditions, and treatment progress with consistency previously only possible in clinical settings. By controlling viewpoint, configuration, and lighting over time, users can create compelling visualizations of subtle changes that occur over weeks and months.

As detailed in our original paper [Tran et al. 2024], our user studies demonstrated significant improvements in geometric accuracy compared to existing tools, while longitudinal case studies show effectiveness in real-world scenarios.

References

Nhan Tran, Ethan Yang, Angelique Taylor, and Abe Davis. 2024. Personal Time-Lapse. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–13.